# UNDERSTANDING OBJECTS & REFERENCES

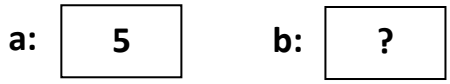
## Some basic facts to be remembered

In Java,

- No variable can ever hold an object. That is, a reference variable can only hold a *reference to an object*.
- Simply declaring a reference variable does ***not*** create an object.
- The **`new`** operator is used to create a ***single object*** and invoke the object's constructor.
- *Several* reference variables can refer to *the same object*. Assigning a reference will ***not*** create a new object
- **Arrays are objects** (*instances of special built-in array classes*). When you create an array, only one object - the array object itself - is created.
- **ArrayLists are objects**. When you create an ArrayList, only one object - the ArrayList object itself - is created. Initially, there are no elements.

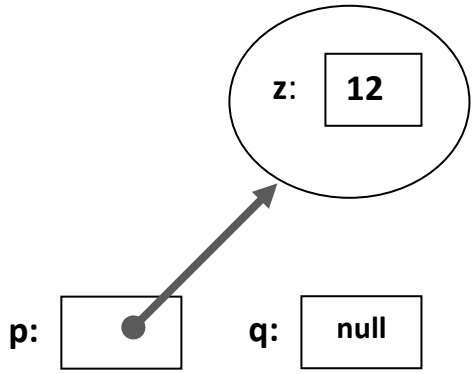
## The four symbols to be used in the drawings

- **VARIABLE**




*In the example, the current value of the variable **a** is **5***
*No value has been assigned to the variable **b** yet.*

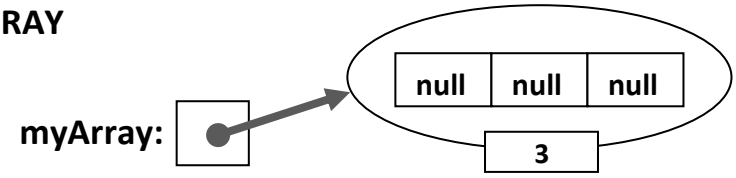
- **OBJECT**
- **REFERENCE**




*In the example, the object has one field called **z**. The current value of **z** is **12**.*

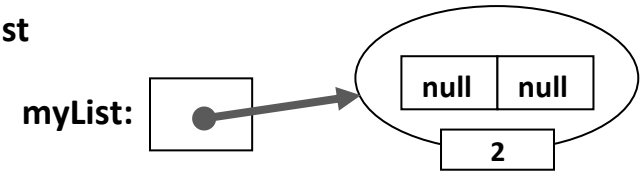
*In the example, the reference variable **p** references to the object above. The reference variable **q** does not reference to any object.*

- **ARRAY**




*In the example, the array has 3 elements. The elements do not hold any references yet. The optional field value (3) indicates that there are 3 elements.*

**ArrayList**




*In the example, the ArrayList has 2 elements. The elements do not hold any references yet.*

```java
class Circle {
    // Fields
    private int r;

    // Constructors
    public Circle() {
        r = 10;
    }

    public Circle(int radius) {
        this.r = radius;
    }

    // Methods
    public int getRadius() {
        return r;
    }

    public void zoom() {
        r = 2 * r;
    }
}
```

```java
class Point {
    private int x;
    private int y;

    public Point() {
        x = 1;
        y = 1;
    }

    public Point(int x, int y) {
        this.x = x;
        this.y = y;
    }

    public void move(int xStep, int yStep) {
        x = x + xStep;
        y = y + yStep;
    }
}
```